Ivanna Dovbysh

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Education

University of Central Florida *August 2025 – May 2027*
MS in Electrical Engineering
- GPA 3.9/4.0 (transcript)

National Technical University of Ukraine *September 2021 - June 2025*
"Igor Sikorsky Kyiv Polytechnic Institute"
BS with Honors in Automation and Instrumentation
- GPA 99/100 (transcript)
- Bachelor's Project: Quadrotor UAV Attitude Estimation System

Research Experience

AirLab *Pittsburgh, PA*
Carnegie Mellon University *May 2026 – November 2026*
Advisor: Prof. Sebastian Scherer
- Developed and evaluated a people-detection pipeline for the DARPA Triage Challenge through large-scale dataset curation, annotation, benchmarking of RF-DETR, YOLO, and GroundingDINO, and automated quantitative evaluation.
- Curated and annotated a casualty-detection dataset of 12,000 frames and 30,000 bounding boxes, now available to the academic community upon request.
- Improved people-detection performance from 0.418 to 0.644 in F0.5 (precision-weighted F-score) through targeted data collection, curation, and model fine-tuning.
- Developed an early-fusion thermal–NIR pipeline for nighttime people detection, reducing false detections caused by shadows and reflections while requiring only a single inference pass.

Learning, Communications, and Wireless Networks Lab *Orlando, FL*
University of Central Florida *September 2025 – May 2027*
Advisor: Prof. Nazanin Rahnavard
- Mentored a team of 6 undergraduate students in developing computer vision applications using Meta Project Aria smart glasses for driver assistance and assistive technologies for visually impaired people.
- Developed a Project Aria–based guidance system for drivers and visually impaired people that guides a moving agent along a desired trajectory while tracking surrounding dynamic objects and predicting their future motion.
- Implemented a real-time pipeline processing RGB and stereo camera data at 20 FPS, using model predictive control to compute collision-aware guidance from the predicted obstacle trajectories.

Intelligent Cyber-Physical Systems (iCPS) Laboratory *Orlando, FL*
University of Central Florida *September 2025 – December 2025*
Advisor: Prof. Truong Nghiem
- Contributed to a loosely coupled SLAM framework integrating 3 sensor types: IMU, 3 RGB-D cameras, and LiDAR, for operating in forested environments with dynamic lighting conditions.

Quadrotor UAV Study *Kyiv, Ukraine*
National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute" *May 2022 – June 2025*
Advisor: Prof. Oleksandr Muraviov
- Developed an attitude estimation algorithm and inertial navigation system for a mini drone, fusing IMU and optical flow data with Extended Kalman and Complementary Filters to enable high-level operator control.
- Designed experiments and analyzed vibration, acoustic, and IMU measurements from 3 types of damaged UAV propellers to develop an autonomous multimodal propeller damage detection system.

LHCb Experiment *Geneva, Switzerland*
CERN *September 2024 – November 2024*
Advisor: Dr. Benjamin Couturier
- Developed an online exercise for CERN LHCb outreach events for high-school students using Python-based web technologies, including PyScript and Pyodide, enabling rapid development and deployment of interactive in-browser Python applications.
- After internship organized CERN lectures for around 200 Ukrainian high school students in cooperation with Vilnius University.

Automatic Control Laboratory *Thunder Bay, Canada*
Lakehead University *May 2024 – August 2024*
Advisor: Prof. Abdelhamid Tayebi
- Contributed to the "Vision-Based Navigation of Quadrotor Drones" project, working with QDrone and Crazyflie platforms, motion capture, and sensor fusion for indoor navigation.

Other Professional Experience

STEM Content Creator *Kyiv, Ukraine*
Self-employed *January 2025 – Present*

- Produced educational content promoting research, engineering, and scholarship opportunities through social media.
- Created educational content in collaboration with the Serhiy Prytula Charity Foundation for both the Foundation's and personal channels.

Physics and Math Tutor *Kyiv, Ukraine*
Akademika, Self-employed *September 2022 – Present*

- Prepared high-school students for national exams (average score 170 on a 100–200 scale) and physics Olympiads (best result: 2nd place in Kyiv).
- Created interactive materials and conducted open lectures on UAV physics for 600 students in 8 schools in Kyiv.

Awards

IROS Video Award *October 2026*

- Awarded for the video "Standards for the Usefulness of AI in Robotics," including a travel grant to the next conference.

Edmund S. Muskie Professional Fellowship *May 2026*

- Full internship funding from the U.S. Department of State; one of 27 fellows selected in 2026.

Fulbright Graduate Student Scholarship *August 2025*

- Full funding for M.S. studies at the University of Central Florida; one of 15 Ukrainian grantees in 2025.

MIRAI Program, Japan *December 2024*

- Academic and cultural exchange program fostering Ukraine–Japan collaboration; one of 2 Ukrainian participants in 2024.

ICT Talent Development: Digital Exploration, China *September 2024*

- Selected for an educational visit focused on technological innovation and industry development in China; one of 3 Ukrainian participants in 2024.

Scholarship of the Verkhovna Rada of Ukraine *September 2024*

- One of the most prestigious national scholarships, awarded annually to 250 undergraduate and graduate students.

Mitacs Globalink Research Internship Program *May 2024*

- Full funding for research at Canadian universities.

Winner, All-Ukrainian Competition of Student Scientific Projects *March 2024*

- Awarded for research on UAV autonomy in a competition among undergraduate and graduate students.

Huawei Seeds for the Future *December 2023*

- Selected participant in Huawei's international technology and cultural exchange program.

Publications

1. **Dovbysh, I. O.**, Muraviov, O. V., Bohdan, H. A., Momot, A. S. (2025). Autonomous UAV navigation: Technologies for orientation and localization. *Applied Issues of Mathematical Modelling*. https://doi.org/10.32782/mathematical-modelling/2025-8-1-5
2. **Dovbysh, I. O.**, Muraviov, O. V. (2023). Dependence of the swarm-leader selection algorithm on the level of UAV autonomy. *Scientific Notes of Taurida National V.I. Vernadsky University, Series: Technical Sciences*, 34(73), 84–90.
3. **Dovbysh, I. O.**, Muraviov, O. V., Galagan, R. M., Bohdan, H. A., Momot, A. S. (2023). Power plants and energy sources of modern UAVs. *Scientific Notes of Taurida National V.I. Vernadsky University, Series: Technical Sciences*, 34(73), No. 5, 16-21.
4. Muraviov, O. V., **Dovbysh, I. O.**, Galagan, R. M., Bohdan, H. A., Momot, A. S. (2023). Development prospects of technologies for increasing autonomy level of UAVs. *Scientific Notes of Taurida National V.I. Vernadsky University, Series: Technical Sciences*, (73) 2, 199–205.
5. Dovbysh, O. S., Bakhtiarova, K. Sh., & Dovbysh, Y. V., & **Dovbysh, I. O.**(2024). Modern aspects of organizing independent work of students of professional pre-higher education institutions during wartime. *Innovative Pedagogy*, 1(70), 18–23.

Conferences

1. **Dovbysh, Ivanna**, & Muraviov, Oleksandr. "Multirotor UAV Propeller Damage Detection Using IMU Measurements." *XVIII All-Ukrainian Scientific and Practical Conference of Students, Postgraduates, and Young Scientists, A Look Into the Future of Instrument Engineering*, Kyiv, Ukraine, May 2025.
2. **Dovbysh, Ivanna**. "Complementary Filter for UAV Attitude Estimation." *XX All-Ukrainian Scientific and Practical Conference of Students, Postgraduates, and Young Scientists, Efficiency and Automation of Engineering Solutions in Instrumentation*, Kyiv, Ukraine, December 2024.
3. **Dovbysh, Ivanna**. "Determination of Damage of Multicopter Propellers Based on the Analysis of Acoustic Signals." *XVII All-Ukrainian Scientific and Practical Conference of Students, Postgraduates, and Young Scientists, A Look Into the*

Future of Instrument Engineering, Kyiv, Ukraine, May 2024.

4. **Dovbysh, Ivanna**. "Methods of Protecting Workers Depending on the Level of Human–Robot Interaction." *XIX All-Ukrainian Scientific and Practical Conference of Students, Postgraduates, and Young Scientists, Efficiency and Automation of Engineering Solutions in Instrumentation*, Kyiv, Ukraine, December 2023.
5. **Dovbysh, Ivanna**. "Issues Related to Using Solar Panels as an Energy Source for UAVs." *XXIV International R&D Online Conference for Students and Emerging Researchers, Science and Technology of the XXI Century*, Kyiv, Ukraine (Online), December 2023.
6. **Dovbysh, Ivanna**. "The Impact of the European Integration Process on the Realization of the Right to Education in Ukraine." *All-Ukrainian Scientific and Practical Conference, Protection of Socially Vulnerable Groups: International and National Dimensions*, Kyiv, Ukraine, November 2023.
7. **Dovbysh, Ivanna**. "Automation of the Thermographic Diagnostic Method for Human Pathologies." *XXII International Scientific and Technical Conference, Instrument Engineering: State and Prospects*, Kyiv, Ukraine, May 2023.
8. **Dovbysh, Ivanna**. "Dependence of Swarm Architecture on the Level of UAV Autonomy." *XVI All-Ukrainian Scientific and Practical Conference of Students, Postgraduates, and Young Scientists, A Look Into the Future of Instrument Engineering*, Kyiv, Ukraine, May 2023.

Service, Outreach & Mentoring

IROS 2026 *September 2026*
- Provided technical support for workshops and conference sessions.

Connecting Champions Mentor *July 2026 - Present*
- Organized one-on-one outreach sessions with children and young adults diagnosed with cancer, introducing them to engineering through engaging, age-appropriate activities.

Fulbright Alumni *April 2026*
- Provided guidance to prospective Fulbrighters through online Q&A sessions and individual feedback on application statements.

Mentoring Youth in Virginia Program *March 2026 - Present*
- Mentored refugees from Afghanistan by providing educational guidance and resources for personal and professional development.

Member of Conference Organization Team *December 2024 - May 2025*
- Organized and hosted XIX All-Ukrainian Scientific and Practical Conference of Students, Postgraduates and Young Scientists "Efficiency and automation of engineering solutions in instrumentation" and XVII Scientific and Practical Conference of Students, Postgraduates and Young Scientists "A Look into the Future of Instrumentation"

Volunteering Teacher *September 2024 - May 2025*
- Provided free mathematics and physics tutoring to high school students from vulnerable groups.

Undergraduate Student Group Leader *September 2021 - June 2025*
- Led and coordinated a group of undergraduate students throughout the bachelor's program, supporting academic progress, communication, and collaborative activities.

Skills

Languages: Ukrainian (Native), Russian (Native), English (C2), German (A2)
Programming Languages: Python, C, C++, Java, JavaScript, PyScript, LabVIEW, MATLAB
Computer Vision & ML: YOLO, RF-DETR, GroundingDINO, OpenCV, PyTorch
Robotics & SLAM: SLAM, sensor fusion, object tracking, trajectory planning, state estimation
CAD & Simulation Systems: SolidWorks, Ansys, Altium Designer, Multisim, Proteus
Hardware & Embedded Systems: Arduino, STM32, Pixhawk, Jetson Nano